# The Pascaline

*West bay 3: Pascal's adding machine (1642) · 19 October 2026*

Babbage's Hall, west bay 3. Blaise Pascal's adding machine (1642), with three digits and his own carry: the **sautoir** ("jumper").

Each digit wheel carries ten pins (the hand turns it by these, like a dial telephone) and, behind them, a snail cam. As the units count from 0 to 9 the cam slowly lifts a weighted lever, the sautoir. As the units pass 9 the cam's step drops away under it, the weight falls, and a pawl on the lever's end kicks the tens wheel on by one pin. On the way up the pawl yields and slips back over the next pin. Because each carry is powered by a weight that was lifted earlier, a carry that ripples (99 + 1 = 100) needs no more force from the hand than any other step. This was Pascal's answer to the problem that wrecked earlier calculators: a long ripple carry jams a machine that drives it straight through gears.

## Running it

Open `pascaline.lua` in bpp, or run it inside the hall (`HALL_START=pascaline`). It starts with a tour: 7, then + 8 = 15 (a carry), + 84 = 99, + 1 = 100 (a double ripple). It prints `RESULT pascaline tour: PASS` when every reading is right.

| Key | |
|---|---|
| `1` `2` `3` | add one to the units, tens, hundreds |
| `R` | reset to 000 |
| `T` | the tour again |
| `Z` | slow motion (1/4 speed) on and off |

Sliders: `friction` (0.15 to 0.45 checked), `turn rate` (how fast the hand turns a digit), `speed`.

Environment, for testing: `PA_TOUR=0` (no tour), `PA_FRICTION`, `PA_RATE`, `PA_DEBUG=1` (prints the digits' values and the levers' and pawls' angles), `PA_EXPORT=1` (rewrites `parts/geometry.scad`).

## How it is built

- **One source.** `pascaline-geometry.lua` holds every moving part's shape as boxes, cylinders and convex hulls, with masses. The simulation builds its bodies from them, and writes them to `parts/geometry.scad`, which `parts/pascaline.scad` builds the printed parts from. What is printed is what was simulated.
- **Bodies.** Each lever and pawl is a compound of its primitives, its origin at its centre of mass. The digits turn on hinges to the frame; the sautoirs pivot on pins in the back plate, at the axles' height; each pawl is hinged to its sautoir, limited so it yields one way only, and held against its stop by a small tail weight.
- **The click.** A real Pascaline has a spring detent on each wheel. Here it is a weak pull toward the nearest whole digit, applied by the hinge's motor.
- **The hand** turns one digit at a time by its hinge's motor, one step per press, then lets go.

## Printing and building

`./make-stls.sh` (needs OpenSCAD, and bpp to refresh `geometry.scad`) writes every part into `stl/`, each turned to lie on the bed. The largest pieces are 230 x 140 mm.

| Part | Print | Size (mm) | Notes |
|---|---|---|---|
| `digit-cam.stl` | 2 | 72 × 83 × 43 | units and tens. Cam on the bed; supports under the pin wheel |
| `digit-last.stl` | 1 | 59 × 59 × 32 | hundreds (no cam) |
| `dial.stl` | 3 | 88 × 88 × 6 | press onto the hub's end; paint the raised numbers |
| `sautoir-0.stl`, `sautoir-1.stl` | 1 each | 215 × 28 × 40/52 | supports under the pawl's carrier arm |
| `pawl-0.stl`, `pawl-1.stl` | 1 each | 31 × 23 × 8 | |
| `frame-front-left/right.stl` | 1 each | 230 × 140 × 6 | the front plate, the digits' bearings and pointers |
| `frame-back-left/right.stl` | 1 each | 230 × 140 × 6 | the back plate, the sautoirs' pivots |
| `frame-base-left/right.stl` | 1 each | 230 × 110 × 8 | |
| `frame-end.stl` | 2 | 58 × 132 × 8 | joins the plates at each end |

Instead of printing the plates you can cut them from 6 mm acrylic (clear, so the works show) or plywood: `stl/cut-front-plate.svg` and `stl/cut-back-plate.svg` are their outlines.

**Also needed**

- 2 steel weights of about **60 g**, each on an **M6 bolt** through the sautoir's weight block (a stack of M6 nuts or washers does). The weight matters: the drop must carry the next wheel a full step.
- 2 pawl tail weights of about **8 g** (an M5 nut glued on). Too light and the pawl swings out instead of catching.
- **3 mm steel rod** for pivot pins: 2 for the sautoirs (into the back plate) and 2 for the pawls.
- Glue or M3 screws for the frame joints; a little dry PTFE or graphite on the cams and pins.

**Assembly.** Glue the plate halves and the base halves. Fit the sautoirs on their pins in the back plate, and the pawls on theirs. Slide each digit's hub, cam first, from behind through the front plate, then press its dial on with 0 at the top under the pointer. Set the units and tens to 0 with each sautoir resting at the bottom of its cam. Join the plates with the end pieces and the base. Then turn the units by its pins and watch the tens step on as it passes 9.

A spring detent on each wheel (a strip of spring steel or a printed leaf pressing on the pins) keeps the digits from being nudged; the simulation's click stands in for it.

See `FINDINGS.md` for how the numbers were arrived at.